

HOSYS - Hotel Billing System

Phase 2 Documentation: Core Restaurant Operations & Billing Workflow

Project Name

HOSYS - Hotel Billing and Restaurant Management System

Technology Stack

- Java 21
 - Spring Boot
 - Spring Data JPA
 - PostgreSQL
 - Maven
 - Hibernate
 - Thunder Client
 - Git & GitHub
-

Phase 2 Objective

The objective of Phase 2 was to implement the complete restaurant operational workflow that connects menu management, tables, orders, billing, payments, and management reporting.

This phase transformed the project from a database-driven application into a fully functional restaurant billing backend.

Modules Implemented

1. Menu Management Module

Purpose

Manage restaurant menu categories and menu items.

Components Created

Menu Category

- MenuCategory Entity
- MenuCategoryRepository
- MenuCategoryService
- MenuCategoryController

Menu Item

- MenuItem Entity
- MenuItemRepository
- MenuItemService
- MenuItemController

APIs Implemented

Menu Category APIs

- POST /api/menu-categories
- GET /api/menu-categories

Menu Item APIs

- POST /api/menu-items
- GET /api/menu-items

Features

- Create menu categories
- Organize menu items by category
- Manage item pricing
- Manage item descriptions
- Track item availability

Example

Category: Main Course

Menu Item: Chicken Biryani

Price: ₹250

Available: Yes

2. Restaurant Table Management Module

Purpose

Manage restaurant tables and their availability.

Components Created

- RestaurantTable Entity
- RestaurantTableRepository
- RestaurantTableService
- RestaurantTableController

APIs Implemented

- POST /api/tables

- GET /api/tables
- GET /api/tables/available

Features

- Create restaurant tables
- Track seating capacity
- Track table availability
- Retrieve available tables

Example

Table T1
Capacity: 4
Status: AVAILABLE

3. Order Management Module

Purpose

Create and manage customer orders.

Components Created

- Order Entity
- OrderRepository
- OrderService
- OrderController

APIs Implemented

- POST /api/orders
- GET /api/orders

Features

- Create customer orders
- Assign orders to tables
- Assign orders to waiters
- Track order status

Example Workflow

Customer sits at Table T1
↓
Waiter creates Order
↓
Order Status = PENDING

4. Order Item Management Module

Purpose

Add food items to customer orders.

Components Created

- OrderItem Entity
- OrderItemRepository
- OrderItemService
- OrderItemController

APIs Implemented

- POST /api/order-items
- GET /api/order-items

Features

- Link menu items to orders
- Store quantity ordered
- Store item pricing
- Calculate item-level totals

Example

Order #1

↓

Chicken Biryani × 2

↓

Price = ₹250 each

5. Billing Module

Purpose

Generate bills from customer orders.

Components Created

- Bill Entity
- BillRepository
- BillService
- BillController

APIs Implemented

- POST /api/bills
- GET /api/bills

Features

- Create bills
- Store subtotal
- Store tax amount
- Store total amount
- Track payment status

Example

Subtotal = ₹500

Tax = ₹25

Total = ₹525

6. Payment Module

Purpose

Manage customer payments.

Components Created

- Payment Entity
- PaymentRepository
- PaymentService
- PaymentController

APIs Implemented

- POST /api/payments
- GET /api/payments

Features

- Record payments
- Store payment method
- Track payment status
- Maintain bill-payment relationship

Example

Bill #1

↓

Payment Method = UPI

Amount = ₹525

Status = SUCCESS

7. Dashboard Module

Purpose

Provide management-level business insights.

Components Created

- DashboardSummary DTO
- DashboardService
- DashboardController

APIs Implemented

- GET /api/dashboard

Metrics Generated

- Total Users
- Total Tables
- Total Orders
- Total Revenue

Example Response

```
{
  "totalUsers": 1,
  "totalTables": 1,
  "totalOrders": 1,
  "totalRevenue": 525
}
```

System Workflow Achieved

The complete restaurant billing flow is now operational.

User (Waiter)

↓

Restaurant Table

↓

Order

↓

Order Item

↓

Menu Item

↓

Bill

↓

Payment
↓
Dashboard Reporting

Database Relationships Implemented

Role
↓
User

MenuCategory
↓
MenuItem

RestaurantTable
↓
Order

User (Waiter)
↓
Order

Order
↓
OrderItem

MenuItem
↓
OrderItem

Order
↓
Bill

Bill
↓
Payment

APIs Successfully Tested

✓ User APIs

✓ Role Seeder

✓ Menu Category APIs

✓ Menu Item APIs

✓ Restaurant Table APIs

✓ Order APIs

✓ Order Item APIs

✓ Bill APIs

✓ Payment APIs

✓ Dashboard APIs

Business Workflow Implemented

Menu Category Creation

↓

Menu Item Creation

↓

Customer Assigned to Table

↓

Order Creation

↓

Order Items Added

↓

Bill Generated

↓

Payment Recorded

↓

Dashboard Updated

Git Milestone

Phase 2 development was committed and pushed to GitHub.

This milestone represents the completion of the Core Restaurant Billing Workflow.

Next Phase

Phase 3 - Security & Enterprise Features

Upcoming Development:

- JWT Authentication
- Login API
- Token Generation
- Token Validation
- Role-Based Access Control
- Admin Authorization
- Manager Authorization
- Waiter Authorization
- Chef Authorization

Future Integrations:

- React Frontend
 - Docker Containerization
 - AWS Deployment
 - Azure DevOps CI/CD Pipeline
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Phase 2 Status

COMPLETED SUCCESSFULLY

The HOSYS backend now supports a complete restaurant billing lifecycle, including menu management, table management, order processing, billing, payment processing, and dashboard reporting.